

$$\mathbb{C}[x]$$

$$f(x) = (x - 1)^2(x + 2)$$

picture of $V(f) \subset \text{Spec}(\mathbb{C}[x])$

A prime ideal is $(x - a)$, so a point is literally a root $a \in \mathbb{C}$.

Quotienting by (f) means: keep only the points where f vanishes.

Repeated factor $(x - 1)^2$ gives a *thickened* point at 1.

$$\mathbb{Z}$$

$$12 = 2^2 \cdot 3$$

picture of $V(12) \subset \text{Spec}(\mathbb{Z})$

A prime ideal is (p) , so a point is a rational prime. (5)

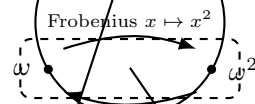
Quotienting by (12) means: keep only the primes dividing 12.

The factor 2^2 gives a thickened point at (2) , just like $(x - 1)^2$ did in $\mathbb{C}[x]$.

$$\mathbb{F}_2[x]$$

$$g(x) = (x + 1)^2(x^2 + x + 1)$$

mult. 2
over $\mathbb{F}_2$: actual roots



over $\mathbb{F}_2$: closed points = irreducible factors

A point of $\text{Spec}(\mathbb{F}_2[x])$ is an irreducible polynomial. Equivalently: a Frobenius-orbit of roots in $\mathbb{F}_2$.

So $(x^2 + x + 1)$ is one point determining even, con-, ω^2 .

Same visual rule in all three worlds:

quotient by an element $\implies$ highlight its prime factors as points.

In $\mathbb{C}[x]$: points = roots $(x - a)$. In $\mathbb{Z}$: points = primes (p) . In $\mathbb{F}_2[x]$: points = irreducibles = Frobenius-orbits of roots.